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A lightweight general PUF framework for resisting machine learning attacks[☆]

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ABSTRACT

Physical Unclonable Function (PUF) is an attractive and low-cost security primitive that requires no storage and is resistant to reverse engineering. However, classical PUFs are highly vulnerable to machine learning attacks, and most attempts to resist these attacks consume excessive resources. To address this challenge, a lightweight general PUF framework is proposed in this paper. Firstly, the framework adopts segmentation processing to introduce structural nonlinearities for the purpose of self-protection. Secondly, the pre-segment response, pre-segment challenges and post-segment challenges undergo XOR processing to introduce challenges obfuscation, which greatly enhances the machine learning resistance of the PUF. In addition, for configurable RO PUF, a novel MUX-based RO (called MRO) is proposed in this paper, which can save resources by 50 %. Implementing a two-segment MRO-MRO instance based on the proposed PUF framework results in reliability, uniformity, and uniqueness that are close to the ideal values. Comprehensive experiments demonstrate that the proposed PUF has the advantages of scalable framework, low resource overhead, and strong resistance to machine learning attacks.

1. Introduction

Physical Unclonable Function is a hardware-based security primitive that utilizes random physical variations introduced during the fabrication process to implement specific pseudo-random sequences, making it a building block to satisfy a wide variety of security applications for resource-constrained devices [1,2]. The design of PUF is not a secret, but its behavior cannot be predicted if there is physical access to the PUF. It is difficult to create two identical PUFs even by utilizing the same fabrication process and hence, it serves as the physical ID of the device. Contrary to non-volatile memories that are susceptible to reverse engineering, PUFs are inherently tamper-resistant and do not need to be stored [3,4]. PUFs can be implemented in field-programmable gate arrays (FPGAs), which are convenient and easy to validate their design.

The input of a PUF is called challenge and its output is called

response, and the combination of input and output is named as the challenge-response pairs (CRPs). PUFs can be categorized into two types: weak PUFs and strong PUFs [5,6]. Weak PUFs have a small number of CRPs. RO PUFs [7], SRAM PUF [8], and DA PUF [9] are typical weak PUFs that can produce reliable CRPs. Due to the limited number of CRPs in weak PUFs, they cannot be the basis for device authentication. On the contrary, strong PUFs are characterized by having a large number of CRPs, the number of which grows exponentially with respect to the input bitwidth, such as Arbiter PUFs (APUFs) [10,11] and BR PUFs [12]. Strong PUFs can be prime candidates for some security applications, such as lightweight authentication, hardware obfuscation [13] and advanced encryption [14,15]. With the development of machine learning, modeling attacks threaten the security of PUFs [16–19]. Attackers try to derive a highly reliable software model to predict the response of the PUF output by collecting the number of CRPs

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and combining it with a suitable machine learning algorithm. Due to the large number of CRPs generated by strong PUF and the spatial correlation between CRPs, strong PUF is highly vulnerable to the modeling attacks. Since modeling attacks are cheap and easy to operate, PUFs resistant to modeling attacks have become a research focus.

Many defense mechanisms have been proposed to resist machine learning attacks. There are three mainstream approaches, which are nonlinear structure, obfuscating the mapping relationship between challenges and response, and cryptographic protection. The purpose of the nonlinear structure is to increase the complexity of the PUF and introduce nonlinearity between challenges and responses. For example, the outputs of multiple APUFs in the XOR APUF are XORed, resulting in nonlinear characteristics. However, the method of achieving randomness by adding nonlinearity to the structure can lead to weak reliability [36]. Also, complex structures are often accompanied by a drain on resources. The obfuscation technique involves masking the challenges or responses of the original PUF, thereby complicating the extraction of authentic CRPs by malicious entities. The FLAM PUF [20] design integrates a LFSR to merge its responses with the LFSR's output, thus continuously concealing the challenges and enhancing security against attacks. The OB PUF [38] framework introduces a streamlined authentication protocol that selectively reveals only the obfuscated elements of the challenges, effectively obscuring the association between the challenges and their corresponding responses. Additionally, MMPUF [21] employs the response from a weak PUF to cloak the overall challenge, further impeding an adversary's ability to access genuine CRPs. While this approach is effective against machine learning attacks, the extra blocks bring about a consumption of hardware resources. Cryptographic protection, such as the AES algorithm [22], employs additional algorithms to protect the PUF. This approach is broadly similar to obfuscation, where protection by introducing extra blocks leads to excessive resource consumption. Since most applications are usually executed on resource-constrained ICs, PUFs need to be as lightweight as possible [23, 24].

In order to solve the problems of weak resistance of strong PUFs to machine learning attacks and high resource consumption of traditional resisting machine learning PUFs, this paper proposes a framework that combines nonlinear structure and obfuscated challenges. The suggested approach entails a multi-stage processing of the PUF, where the PUF linked to the multiplexer's (MUX) selection terminal acts as the front stage. Subsequently, the two PUFs connected to the MUX inputs constitute the next stage, with the response from the front stage determining the selection of one PUF in the following stage to generate the output. The nonlinear structure is realized by selecting the response of either of the two PUFs of the next stage as the output through the response of the front stage of the multiplexer. In this research, the obfuscation of challenges is achieved through the creation of an obfuscation unit, comprised of XOR gates that introduce dissimilarity, which we have named CXOR. The CXOR is adeptly designed to uniquely transform the challenge of the challenge of the front stage, the response from the front stage, and the challenge of the next stage, thereby effectively obfuscating the challenges. The simple structure of the proposed framework and the simple heterodyne logic unit used in the obfuscation block significantly reduces the consumption of hardware resources. The main contributions of this article can be concluded as follows.

- A framework that combines nonlinear structures as well as obfuscated challenge methods, is proposed to improve the capability of the classical strong PUF resistance to machine learning attacks. The proposed method employs APUF, MRO, and MPUF [43] to build six different structures for evaluation, i.e., MRO-MRO, APUF-MRO, APUF-APUF, APUF-MPUF, MPUF-MPUF, and 3-APUF, with prediction rates close to the desired value of 50 % under LR, CMA-ES, and ANN tests. The experiments demonstrate that the framework can enhance the machine learning resistance for different strong PUFs.

- For configurable RO PUF (CRO-PUF), a novel MUX-based RO (MRO) structure is proposed, and a two-segment MRO-MRO structure is implemented based on the proposed PUF framework, which consumes fewer resources than all previous works.
- The framework proposed in this paper can be processed in stages according to the designer's requirements, which means the framework is easy to scale. According to the segmentation characteristics, it can be categorized into a two-stage processing framework, a three-stage processing framework, or even more complex configurations.
- For the proposed MRO-MRO structure, its performance is tested on Xilinx Virtex-7 and Xilinx Spartan-6 and the final results show that the uniformity, uniqueness and reliability of the structure are close to the desired values.

The remainder of this paper is organized as follows. The background and related work are detailed in Section II. Section III describes the proposed framework and the RO deployment of the framework. Experimental results and analysis as well as the proof of generalization are illustrated in Section IV. Finally, Section V concludes the paper.

2. Related work

This section presents traditional CRO PUF, APUF, three machine learning attack algorithms, and methods to strengthen resistance modeling against them. These are the relevant foundations for introducing the MRO and the framework of resistance machine learning algorithms proposed later.

The concept of RO PUF was first proposed by G. Edward [7], as shown in Fig. 1(a), consisting of a pair of ring oscillator chains, where a single ring consists of an odd number of inverters. The RO PUF compares the number of oscillations between the two rings to generate a one-bit response. This structure has good performance, but it is unable to generate more CRPs, and consumes more resources. In order to solve this problem, many articles based on the oscillating ring start to work on changing the original ring in the non-gate, to take the heterodyne gate

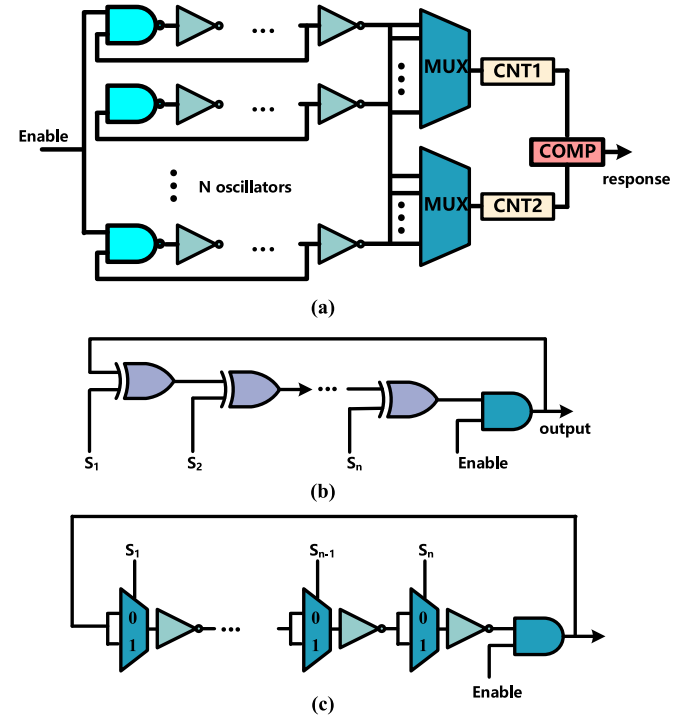


Fig. 1. Related works on Configurable RO PUF: (a) RO PUF, (b) Ring composed of hetero-or gates, and (c) RRO.

A Configurable RO PUF

combined with the data selector and other ways to design to improve the hardware utilization of the overall PUF. This kind of ring can be configured by challenges in a way that can reach an exponential level, so the PUF composed of configurable rings is a strong PUF. Zhang et al. [25] proposed an XCRO PUF that uses a hetero-or gate to replace an inverter, and this structure increases configurability and reduces the consumption of hardware resources, as shown in Fig. 1(b). In order to further simplify hardware complexity, Cui et al. [26] uses a hetero-or gate to replace the inverter to propose an XCRO PUF with a reconfigurable RO (RRO), in which only one inverter and one multiplexer are used for each delay unit. As shown in Fig. 1(c), this reconfigurable design approach greatly expands the number of CRPs and thus achieving excellent uniqueness and reliability, but machine learning can easily defeat the structure.

B. Arbiter PUF

APUF is one of the most popular and well-studied PUFs, as shown in Fig. 2. An n -bit APUF is composed of n cascaded MUX pairs and an arbiter at the end. The arbiter, which can be either a DFF or a combination of two NAND gates, determines the parallel path or cross-path of each MUX pair through challenges. Although the APUF is structurally symmetric, it impacts the propagation speed of the upper and lower paths due to inherent variations introduced during the manufacturing process. After the path is configured by the challenge, a rising edge pulse is given at the input, which determines whether the response at the output is a logic "1" or a logic "0" due to the delay difference between the upper and lower paths.

The operation of the arbitrated PUF can be described by a linear delay model [23], where the cumulative sum of each stage constitutes the total delay of the step signal. The final delay can be defined as

$$\Delta = \vec{\omega}^T \vec{\phi} \quad (1)$$

where $\vec{\omega}$ is the delay vector at each level of the APUF induced by the process variation and $\vec{\phi}$ is a function of the n -bit challenge $\vec{\omega}$. $\vec{\omega}$ can be expressed as

$$\vec{\omega} = (\omega_1, \omega_2, \dots, \omega_{n+1})^T \quad (2)$$

where $\omega_1 = [(\delta_1^0 - \delta_1^1)/2]$, $\omega_i = [(\delta_{i-1}^0 + \delta_{i-1}^1 + \delta_i^0 - \delta_i^1)/2]$, for all $i = 2, \dots, k$, and $\omega_{n+1} = [(\delta_n^0 - \delta_n^1)/2]$. Among them, $\delta_i^{0/1}$ is the runtime delay in the i th stage for the parallel and crossed signal path, respectively. Furthermore

$$\vec{\phi}(\vec{C}) = (\phi_1(\vec{C}), \dots, \phi_n(\vec{C}), 1)^T \quad (3)$$

where $\vec{\phi}_l(\vec{C}) = \prod_{i=1}^n C_i$ for $C_i \in \{-1, 1\}$, $l = 1, \dots, n$.

If $\Delta > 0$, the response bit r will be 1. Otherwise, r will be 0. The goal of machine learning attacks is to find the most appropriate vector $\vec{\omega}$ through $\vec{\omega}^T \vec{\phi} = 0$ to determines a separate hyperplane.

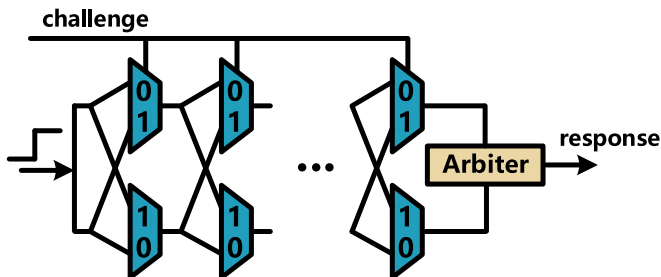


Fig. 2. The structure of the Arbiter PUF.

C. Machine Learning Attacks

- 1) *Logistic Regression (LR)*: LR can be utilized to address binary classification problems and was the first ML algorithm employed to counter PUF. The optimal coefficients of the prediction function are solved by constructing the prediction function, loss function and minimizing the loss function. Logistic regression and linear regression by many of the same places, the main difference is that the multiple linear regression uses the form of $y = \omega^T x$, while the logistic function uses the sigmoid function as the discriminant function, the expression is shown below.

$$y = \frac{1}{1 + e^{-z}} \quad (4)$$

where $z = \omega^T x$. this form of discriminant function maps the input range of the function from $(-\infty, +\infty)$ to the interval $(0, 1)$, so the logistic regression is often used to solve binary classification problems. LR is a linear classification model, and it is effective in attacking PUFs with a linear structure [27,28]. In our study, we employed the open-source Python implementation of RProp LR, developed by Ulrich et al. [16], which is accessible via [46].

- 2) *Covariance Matrix Adaptation Evolution Strategy (CMA-ES)*: Evolutionary Strategies (ES) is another powerful ML technique that primarily uses mutation and selection to model strong PUFs. The main idea is to generate random PUF instances and iteratively find the one that best matches the real PUF model. The reliability-based CMA-ES [19] attack utilizes repeated measurements to observe the reliability of response bits. The reliability information is then fed into the fitness function to find the most suitable delay parameter. It outperforms conventional modeling attacks on heterogeneous APUFs with a large number of heterogeneous APUFs. It is shown in the literature [29] that even highly confusing responses can be attacked by CMA-ES. For the reliability-based CMA-ES algorithm, we adhere to Becker's methodology [19] and employ the default parameters as specified in Ref. [45].
- 3) *Artificial Neural Network (ANN)*: ANN is a network structure consisting of connected artificial neural units that can model complex relationships between inputs and outputs using computational and statistical data modeling tools. Neural networks are composed of different layers, which are called as input, output and hidden layers. The first layer of the network that receives inputs is called the input layer while the last layer of the network is called the output layer. The middle layer is known as hidden layer. ANNs offer more capabilities in capturing high dimensional relationships between the input and output spaces as compared to conventional techniques. Artificial neural networks achieve excellent performance against PUFs with additional nonlinearities. For example, ANN can effectively predict Feed-Forward PUF (FF-APUF) [30]. In this study, we employed two modeling attacks as proposed in the literature [43] to challenge our proposed architecture, named logical approximation and global approximation. These techniques utilize ANNs to delineate the nonlinear structure of the PUF. The logical approximation approach applies linear functions to approximate logical operations, constructing an accurate soft model predicated on the amalgamation of logic gates within the PUF. Conversely, the global approximation strategy adopts the sinc function, noted for its filtering capabilities, to accurately model the mapping relationship between the challenge and response mechanisms.

D. PUF Security Enhancement Methods

- 1) *Structural Nonlinearity*: this approach starts from the PUF structure itself and adds nonlinearities to the original PUF structure, including the addition of feedback loops or multipath selection, etc. FF-APUF introduces nonlinearities by adding feedback loops but has been shown to be insufficient to resist ML attacks [16]. The rest of the nonlinear work such as XOR APUF, cascaded PUF [31], MPUF [42],

- etc. add nonlinearity for the purpose of resisting machine learning attacks. However, most of these structures are vulnerable when encountering advanced attack methods. If the nonlinearity of the structure is introduced in a transitional manner, it may lead to the emergence of unreliability and excessive resource consumption of the final PUF.
- 2) **Obfuscation:** challenges or response obfuscation is used to prevent an attacker from collecting valid CRPs for modeling attacks by hiding the relationship between the challenges and the response. Obfuscation methods include combinatorial methods, random bit methods, cryptographic methods and heterogeneous gate methods. Combination method is to combine two or more PUFs and use the output of one PUF as the input of another PUF, such as MMPUF [21]. Random-bit methods are used to achieve resistance to machine learning attacks by introducing random numbers to randomize the challenge or response of a PUF, such as RPUF [32]. Cryptographic methods such as Chaotic-Bistable Ring PUF [33], achieve obfuscation of the response of BR PUF by introducing a one-dimensional random function. Obfuscation can be effective against machine learning attacks, however, the introduction of extra blocks usually brings resource consumption.
 - 3) **Cryptographic Protection:** Cryptography is a method that combines traditional encryption algorithms with PUF to improve security. The cryptographic algorithms used for PUF usually are hash function, AES, matrix encryption and so on, in which hash function is the most popular. Controlled PUF [34] utilizes hash functions to obfuscate the challenge and response in order to prevent selective attacks on the challenge and to reduce the correlation of the response. In addition to hash algorithms, there are also algorithms such as AES algorithms to protect circuits [22]. These methods are similar to the obfuscation method in that they resist machine learning attacks by introducing extra blocks, which brings the problem of high resource consumption and does not fulfill the lightweight requirement of PUF design.

3. Proposed framework

PUFs are usually applied in resource-constrained ICs. As mentioned above, the introduction of nonlinearities in the transition will lead to lower reliability, while the overly complex structure also leads to resource overload. Introducing additional blocks for obfuscation does not protect the PUF itself and also leads to resource constraints. Similarly, encryption algorithms will also bring more hardware resource consumption. Considering the requirement that PUF should be lightweight and can be put into applications, in this paper, we will use an approach that combines nonlinear structures as well as obfuscation to design a lightweight general-purpose resisting machine learning framework. The framework takes into account the resource consumption on the whole, so the proposed structure is not complex, and the obfuscation only uses simple XOR logic and only partially motivates obfuscation.

In this section, we will illustrate the principle of the proposed framework.

This framework employs a nonlinear structure as well as obfuscation of partial challenges to achieve resisting machine learning. The framework processes the whole PUF in stages and introduces a lightweight obfuscation block CXOR to obfuscate the partial challenges. The block diagram is shown in Fig. 3.

In the two-stage processing framework, C1 and C2 are the adjacent stages of the challenge. For example, if the whole challenge is 64-bit, C1 is the first 32-bit challenge and C2 is the last 32-bit challenge. PUF1, PUF2, and PUF3 are three different strong PUFs, and C1 is directly inputted to PUF1 to get the response of R1. Specifically, R1 has two functions here, the first one is to select either PUF2 or PUF3 in the second stage of the processing as the output of the response, and the second one is to be inputted to the CXOR obfuscation block to participate in the obfuscation computation. Within the entire framework, the

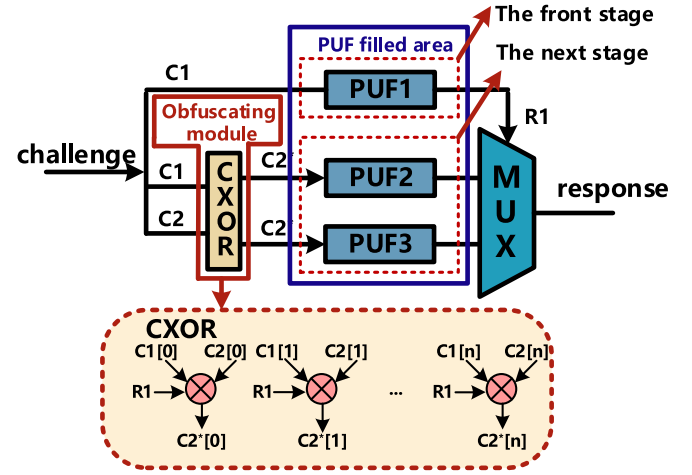


Fig. 3. Block diagram of the proposed structure with two-stage processing.

A Nonlinear Obfuscation Framework

response is not exclusively derived from a solitary PUF, thereby impeding the attacker's ability to model the PUF to a certain degree. Regarding the two-stage processing framework, the response emerges from the collaborative influence of three internal PUFs and collectively defies approximation through a linear function. The overall structure of staged processing with outputs from different PUFs introduces nonlinearity in the structure.

The CXOR unit consists mainly of XOR, receiving inputs from the challenge C1 in the first stage, the challenge C2 in the second stage, and R1, which is the output obtained from PUF1 in the first stage. The overall structure is depicted in Fig. 3. Each bit of the challenge C1 and each bit of the challenge C2 are synchronized with R1 to obtain the processed C2*. The result of C2* is related to the inputs of C1, C2, and R1. According to Table 1, C2* has eight possible values. When C1 and C2 are determined, C2* has two possible results, which is determined by the output of PUF1 (R1) but cannot be predicted in advance, resulting in that the final input C2* stimulus to PUF2 and PUF3 is also unpredictable.

To ensure the proposed framework's applicability, it is essential to first verify that the final response obtained is unbiased. The framework is required to be populated by strong PUFs and the ideal value of uniformity of strong PUFs is 50 %. The final output response R result probability formula after MUX is expressed as follows:

$$P(R) = P(R_1 R_2 + \bar{R}_1 \bar{R}_3) \\ = P(R_1 R_2) + P(\bar{R}_1 \bar{R}_3) - P((R_1 R_2) \cap (\bar{R}_1 \bar{R}_3)) \quad (5)$$

Where $R_1 \sim R_3$ is the response output by PUF1~3 respectively. PUF1 cannot support R1 select 0 and 1 at the same time, where $P((R_1 R_2) \cap (\bar{R}_1 \bar{R}_3)) = \emptyset$. Here it is assumed that PUF1, PUF2, and PUF3 satisfy the homogeneity of 50 %, and because these three items are independent of each other, the formula can be further expressed as

Table 1
CXOR truth table.

R1	C1	C2	C2*
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$\begin{aligned} P(R) &= P(R_1 R_2) + P(\overline{R_1} R_3) \\ &= P(R_1)P(R_2) + P(\overline{R_1})P(R_3) \\ &= 50\% \times 50\% + 50\% \times 50\% = 50\% \end{aligned} \quad (6)$$

From the above equation, it can be seen that the proposed framework does not affect the uniformity, making the design feasible. In addition, the proposed framework can be modified flexibly, but it follows the mechanism that the response of the previous stage selects either one of the two PUFs in the next stage, and then CXOR confuses the two PUFs input to the next stage. A three-stage processing framework is shown in Fig. 4, and a four- or more-stage framework can be designed in the same way.

Prior to the introduction of the configurable Ring Oscillator (RO), RO was composed of an odd number of inverters. By comparing the oscillation times of two different rings simultaneously, a stable response could be obtained, but at the cost of excessive resource consumption. The work of CRO is presented in Section II-A, and its main purpose is to save resources and achieve the requirement of lightweight PUF. This paper designs a new RO loop, which belongs to CRO. In contrast to previous designs, this study employs only a NAND gate and several four-input selectors to construct a new loop for the oscillating PUF, thereby meeting the criteria for resource efficiency.

The LUT serves as the fundamental component in an FPGA, designed to store logic functions. It usually consists of a small memory and a data selector. The memory cell in the LUT uses a lookup table to store function values. Each input combination corresponds to an output value, and the corresponding output can be quickly found by a lookup table. The four-input selector can be configured by changing the parameters of the six-input LUT and forming a configurable ring with respect to multiple four-input MUXs connected to NAND, which is referred to as MRO in this paper. Since a single LUT in the MRO can control two bits of PUF challenge, and the entire loop utilizes only one NAND gate and a four-input MUX, it significantly reduces resource usage. In contrast, conventional configurable loops, such as the RRO shown in Fig. 1(c), require one LUT per bit along with additional gate circuits. As a result, the MRO achieves at least a 50 % reduction in resource utilization. The overall structure of MRO is shown in Fig. 5.

C. MRO-MRO Design

In the oscillating class of PUFs, a counter needs to be employed to count the number of oscillations. Two CROs with the same structure are configured with the same challenge, and the final response is obtained by comparing the number of oscillations of the two rings through a comparator. Deploying PUFs with CROs requires a significant amount of resources, necessitating six counters and three comparators to process just two stages of the frame. To conserve resources, this design introduces a method for oscillating PUF applications that minimizes resource usage. Specifically, it employs MUXs in conjunction with demultiplexers (DEMUXs) to enable a single pair of counters and one

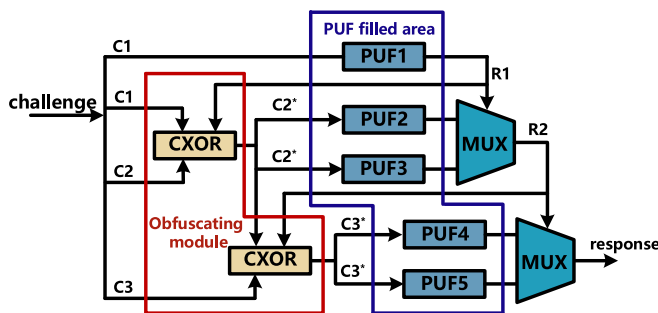


Fig. 4. Block diagram of the proposed structure with three-stage processing.

B. MRO Design

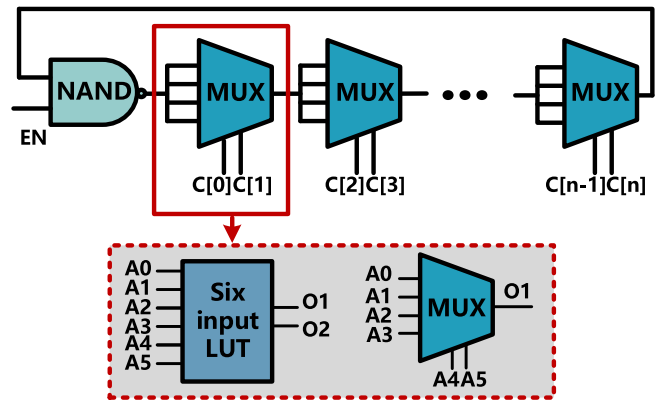


Fig. 5. MRO structural diagram.

comparator to handle multiple rings. Incorporating the previously mentioned MRO, the two-stage MRO-MRO PUF configuration is illustrated in Fig. 6.

In this design, six MROs as introduced in III-B are used, where CNT1 and CNT2 are counters and COMP is a comparator. MRO1 and MRO1* together form a pair of rings for the oscillating PUF1, while the remaining four rings are allocated to PUF2 and PUF3, respectively. Firstly, the whole challenge is divided into C1 and C2 of the same bit width, where C1 is fed to the two rings of PUF1 and C2* is fed to the four rings of PUF2 and PUF3 in the second stage. The initial value of *sel* is 0. In this case, the channel 0 of the first pair of MUX and DEMUX is selected, and at this time the PUF1 consisting of MRO1 and MRO1* as well as the counters and the comparators works, and finally gets the R1. The result R1 is input to DFF through channel 0 of DEMUX, and here DFF plays the role of saving R1. Secondly, a rising edge is input to the *en_dff* after obtaining R1, then the data is input to the Q port at the D terminal, which in turn connects the CXOR obfuscation block with the second pair of MUXs. And at this stage, R1 is involved in the selection of the second PUF and the generation of the C2* process. Finally, the value of *sel* is configured as 1 for the ring of the second stage selected PUF is connected to the counter and comparator and at the same time the channel 1 port of the DEMUX is opened and the final response is obtained after a period.

Since the ring is firstly proposed in this work and it is the basis of the designed MRO-MRO structure, enough performance evaluation would be conducted.

4. Experimental results

The framework was originally designed to achieve resistance to modeling attacks while keeping resource consumption efficient. For this purpose, a lot of work has been done to optimize hardware resources. The consumption of resources is particularly noticeable on the oscillating class of PUFs, so an efficient MRO is designed and a method only

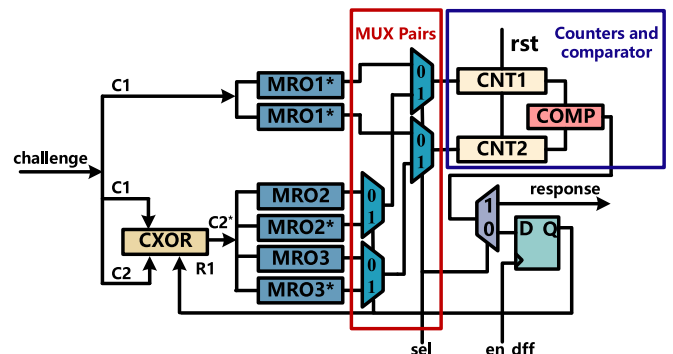


Fig. 6. MRO implementation circuit structure diagram.

using a pair of counters and a comparator is proposed for the two-stage processing structure. This section provides an experimental analysis of the MRO-MRO structure and a proof of generalization of the classical strong PUF application in the proposed PUFs structure.

A. Uniformity

Consistency in a PUF response is determined by the balance of 1's and 0's. A significant imbalance allows an attacker to more easily predict the response. In fact, if the "1/0" ratio is 50 %, the consistency of the PUF is ideal. The consistency of the PUF can be measured by the hamming weight (HW) of the response sequence, which is defined by the following formula.

$$Uniformity = \frac{1}{n} \sum_{i=1}^n (R_i) \times 100\% \quad (7)$$

where n denotes n bits and R_i represents the i th response.

Ten MRO-MRO structures built by MRO were implemented in Xilinx Virtex-7 devices, and ten PUFs were implemented across two Spartan-6 boards. Consistency was achieved at a room temperature of 25 °C. The reference voltage for the Xilinx Virtex-7 was set to 1.0V, and for the Xilinx Spartan-6, it was set to 1.25V.

Fig. 7 displays the uniformity histogram of the proposed structure. On the Virtex-7 platform, the average uniformity is 47.46 % with a standard deviation of 5 %. On the Spartan-6 series, the average uniformity is 51.26 % with a standard deviation of 4.7 %. The uniformity in both experiments approaches the ideal value of 50 %.

B. Uniqueness

Uniqueness is a measure of inter-chip variation by evaluating the difference among m different designs of a particular PUF circuit design. Ideally, when the same PUF circuit is implemented on multiple devices, it should yield about 50 % of the average Hamming distance (HD) when comparing the response produced by the same challenge. Ideally, the response produced by the same challenge on different devices should obey a binomial distribution. The uniqueness of a device is defined as follows:

$$Uniqueness = \frac{2}{m(m-1)} \sum_{i=1}^{m-1} \sum_{j=i+1}^m \frac{HD(R_i, R_j)}{n} \times 100\% \quad (8)$$

Equation (8) is the average of all possible pairwise averaged HDs from m devices, where m denotes the number of realized PUFs, R_i and R_j represent the responses of two non-identical PUFs, respectively. The histograms of the uniqueness from the MRO-MRO experiments tested on Virtex-7 and Spartan-6 are shown in Fig. 8(a) and (b), respectively. The average uniqueness achieved on Virtex-7 is 50.48 % with a standard deviation of 4.03 %; on Spartan-6, the average uniqueness is 50.11 % with a standard deviation of 4.76 %. On both experimental platforms,

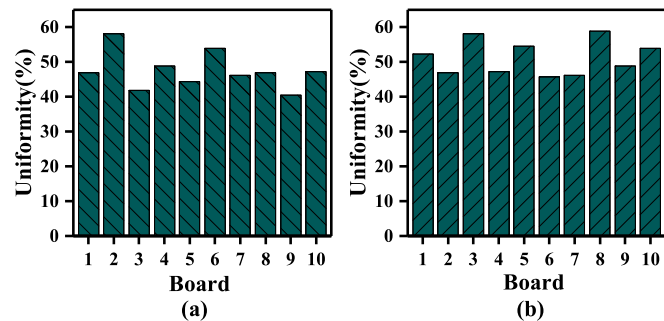


Fig. 7. The uniformity result of the MRO-MRO design: (a) Virtex-7 platform and (b) Spartan-6 platform.

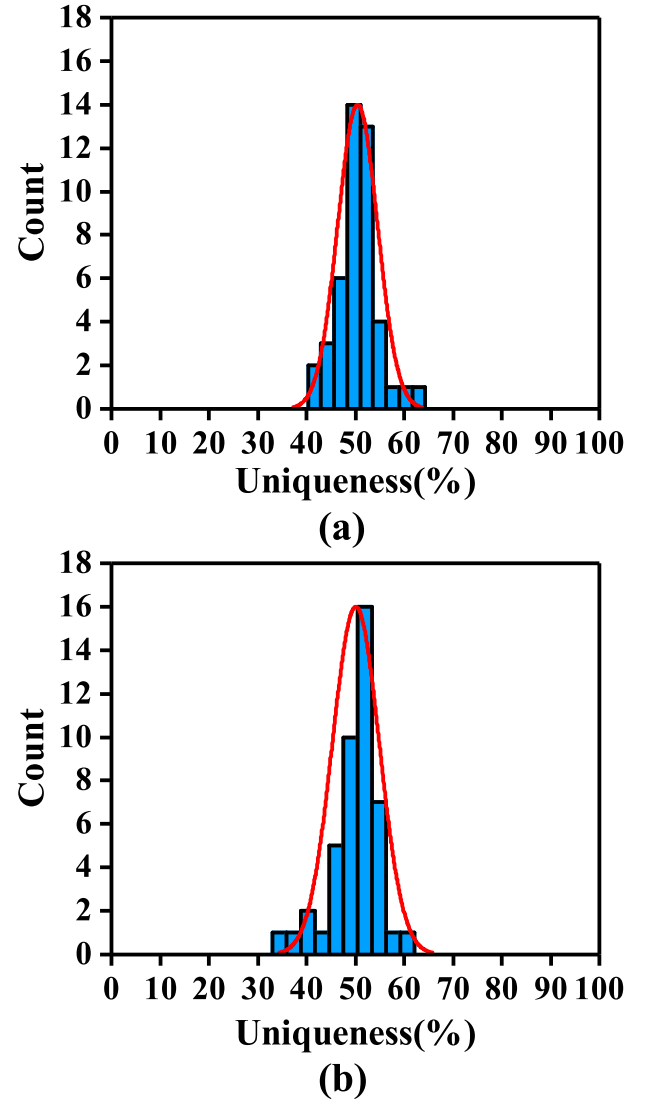


Fig. 8. The uniqueness result of the MRO-MRO design: (a) Virtex-7 platform and (b) Spartan-6 platform.

C. Reliability

the uniqueness closely approaches the ideal value of 50 %.

Ideally, a PUF within a device should reproduce its output perfectly. However, environmental factors such as temperature and voltage fluctuations can affect the PUF's output, necessitating a reliability metric to assess the PUF design's consistency in reproducing responses. For a device, reliability is measured by finding n response samples under baseline conditions and comparing them to the n responses under baseline conditions under different operating conditions to obtain the average on-chip HD, which is defined as follows:

$$HD_{intra} = \frac{1}{m} \sum_{t=1}^m \frac{HD(R_i, R'_{i,t})}{n} \times 100\% \quad (9)$$

where $R_{i,t}$ is the first sample in different environments, and the expression of reliability is defined follows:

$$Reliability = 100\% - HD_{intra} \quad (10)$$

During reliability tests, 8000 responses were collected under the same challenge at different temperatures ranging from 0 to 80 °C and tested on Virtex-7 and Spartan-6. The reliability test for Virtex-7 was conducted at 1.0V voltage and a temperature of 25 °C, with the

histogram shown in Fig. 9(a). The measured reliability was 96.13 % with a standard deviation of 0.76 %. Fig. 9(b) displays the histogram for the Spartan-6 platform at a reference voltage of 1.25V, achieving an average reliability of 97.35 % with a standard deviation of 0.98 %.

To assess reliability under voltage variations, 8000 CRPs were compared. The entire experiment was conducted at 25 °C, with the key difference being the default voltages for Virtex-7 and Spartan-6. The voltage experiments for both platforms ranged $\pm 10\%$, with the dot plots shown in Fig. 10. Fig. 10(a) illustrates the dot plot for Virtex-7, achieving a reliability of 96.80 % with a standard deviation of 1.28 %. Fig. 10(b) displays the dot plot for Spartan-6, where the voltage reliability reached 96.77 % with a standard deviation of 1.17 %.

Table 2 gives the amount of resources required to implement the 128-bit MRO-MRO architectures. The 128-bit MRO-MRO requires only 268 LUTs and 122 DFFs. MRO-MRO consumes fewer resources than all the previous works and does not require RAM cells. The enhanced efficiency of the MRO arises from its markedly reduced reliance on LUTs compared to the traditional CRO for identical bit widths, with the added benefit that the four-input MUX can be configured using just two bits. Moreover, the MRO's design excludes any superfluous gate structures, resulting in a resource conservation that is nearly twice as efficient as previous models. Comprehensive optimization also takes into account resource consumption; the trio of PUFs integrated with the MRO necessitates merely a pair of counters and a single comparator, thereby achieving an optimized allocation of resources. We replicated the experiments of Chaotic PUF [33], RSO PUF [36], CA PUF [40], and FF-APUF [41] on the Virtex-7 platform using identical methodologies for a comparative analysis. Table III shows the performance comparison of the realized MRO-MRO structure with the previous research results, illustrating that the proposed MRO-MRO structure exhibits superior performance and higher reliability under voltage experiments as compared to the previous studies. This reliability is partly attributed to the fact that MRO-MRO uses only a two-phase framework, with fewer effects of randomness introduced by the structure and a simple structure, resulting in a near-ideal reliability. The entire experiment was conducted using MicroBlaze to collect data from the designed PUF circuit, with the CRPs being gathered through the serial port. The power consumption was simulated using Vivado, and the total power

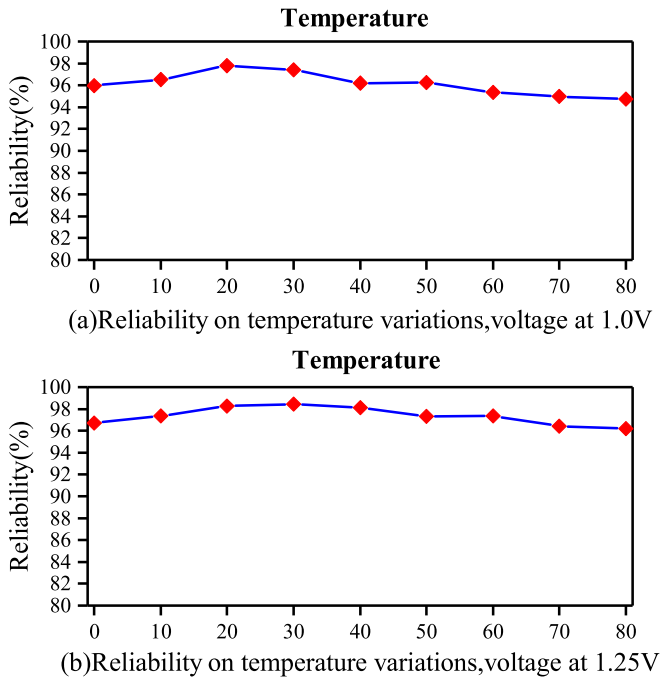


Fig. 9. Reliability under temperature experiments (a) Virtex-7 platform and (b) Spartan-6 platform.

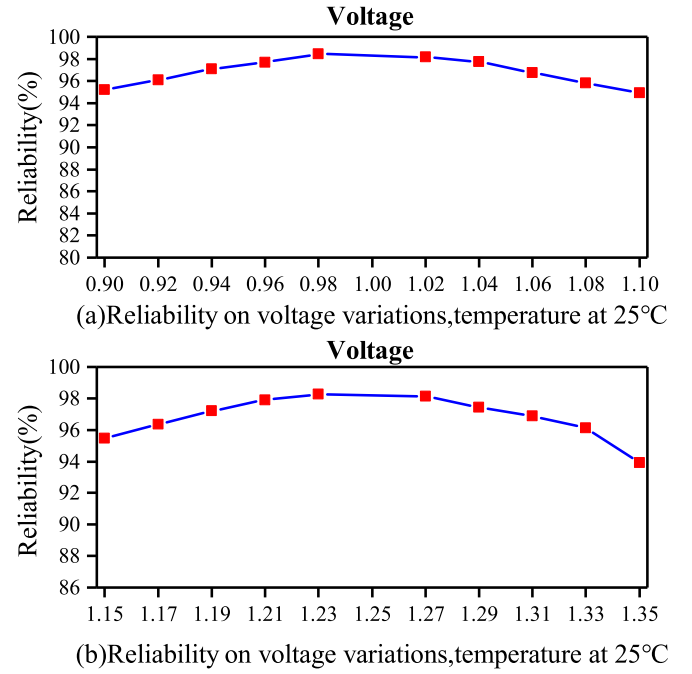


Fig. 10. Reliability under voltage experiments (a) Virtex-7 platform and (b) Spartan-6 platform.

TABLE II

Resource comparison.

Methods	LUTs	DFFs	RAM
PUF-FSM [35]	960	1500	N/A
Chaotic PUF [33]	1444	926	N/A
RSO PUF [36]	395	176	4 KB
CT PUF [37]	731	486	N/A
Dual-mode PUF [28]	678	353	N/A
RPUF [32]	350	389	N/A
OB-PUF [38]	680	360	N/A
Slender PUF [39]	1168	1400	4 KB
Generic Framework [44]	306	298	N/A
This work	268	122	N/A

TABLE III

Performance comparison.

Methods	Uniqueness Original	Virtex- 7	Reliability Original	Virtex- 7	Uniformity Original	Virtex- 7
Chaotic PUF [33]	48 %	49.81 %	N/A	N/A	48.5 %	48.61 %
RSO PUF [36]	50.1 %	50.52 %	N/A	96.15 % (T) 96.20 % (V)	N/A	47.16 %
CA PUF _{AP} [40]	47.03 %	48.46 %	88.47 %	91.36 % (T) 88.63 % (V)	46.38 %	51.89 %
FF-APUF [41]	41.53 %	40.85 %	97.10 % (T) 93.90 % (V)	97.47 % (T) 93.95 % (V)	49.73 %	47.74 %
This work (Virtex-7)	50.48 %		96.13 % (T) 96.80 % (V)		47.46 %	
This work (Spartan-6)	50.11 %		97.35 % (T) 96.77 % (V)		51.26 %	

consumption for the provided example experiment was found to be 0.386 W, with a dynamic power consumption of 0.142 W. The total power consumption of the designed MRO-MRO circuit was 0.003 W, accounting for only 1 % of the total power consumption in the experiment.

D Modelling Attacks Test

In order to evaluate the resistance to machine learning attacks of the proposed two-level framework MRO-MRO, APUF, XOR APUF [7], and MPUF with a nonlinear structure are used here for experimental comparison. APUF can be easily defeated by linear algorithms, while XOR APUF is resistant to modeling, but recent work has used approximation attacks to successfully breach the structure [43]. In this paper, LR, CMA-ES, and ANN methods are used for evaluation. The final obtained results are shown in Fig. 11, where APUF is almost 100 % compromised and is not at all resistant to these three algorithms. For XOR APUF, the prediction rate approaches 100 % under CMA-ES algorithm attacks. With 35,000 CRPs, the ANN algorithm can completely break through 2-XOR-APUF. For 3-XOR-APUF, the prediction rate exceeds 90 % with 25,000 CRPs, making the PUF no longer secure. MPUF possesses excellent resistance against both algorithms LR and CMA-ES but possesses low resistance against ANN. The MRO-MRO structure proposed in this work possesses best resistance to modeling attacks, with its prediction rate close to the ideal value of 50 % under the attacks of these three algorithms.

We referenced previous works on machine learning resistance and used one million CRPs, employing the data from the original paper. The experimental results are presented in Table 4. Notably, while maintaining resistance to machine learning attacks, the resource consumption of the MRO-MRO design is lower compared to Chaotic PUF, RSO PUF, and the Generic Framework.

E. Proof of Generality

The main purpose of the proposed PUF is to strengthen the resistance to machine learning attacks while saving resources. In addition to the application of MRO, it is also necessary to enhance the resistance to modeling attacks on the traditional strong PUF. Here, APUF, the PUF designed with MRO, and the combination of MPUF with the proposed PUF are selected for comparison, finally APUF-APUF, APUF-MRO, APUF-MPUF, MPUF-MPUF, and 3-APUF are implemented, respectively. APUF-APUF represents a two-stage APUF built by the proposed PUF, where both the first stage and second stage employ the APUF. APUF-MRO is the PUF with APUF in the first stage, and the second stage is composed of MRO rings. Similarly, APUF-MPUF as well as MPUF-MPUF are processed identically in the first stage as well as in the second stage. The proposed PUFs with two processing stages are constructed using a

TABLE IV
Machine learning resistance comparison.

Methods	CRPs	Prediction accuracy		
		LR	CMA-ES	ANN
APUF	5×10^3	100 %	99.6 %	100 %
MMPUF [21]	5×10^4	55.39 %	71.96 %	N/A
Chaotic PUF [33]	2×10^4	53.85 %	N/A	N/A
CT PUF [37]	10^6	>55 %	>60 %	>50 %
RSO PUF [36]	10^5	53.79 %	>53 %	>53 %
OB-PUF [38]	2×10^4	63.27 %	N/A	N/A
Generic Framework [44]	7.2×10^5	53 %	N/A	N/A
This work	10^6	52.73 %	55.91 %	56.13 %

two-stage approach, while 3-APUF represents the three-stage PUF for APUF application, whose framework is shown in Fig. 3. And these five PUFs are evaluated by machine learning attacks, and three algorithms, LR, CMA-ES, and ANN, are used for the attack. The results of machine learning are shown in Fig. 12, in which the machine learning prediction rates for all five structures are close to the ideal value of 50 %. It should be noted that this section is only to verify the proposed PUF can enhance the machine learning capability of classical strong PUF, and the resource requirements are not considered.

When using configurable rings for framework filling, the combination of MUX and DEMUX can be employed, which helps reduce the number of counters and comparators, as demonstrated in the MRO-MRO instance. For delay-based PUFs, the framework can be directly filled, as they do not require counters or comparators, resulting in relatively lower resource usage. Due to the integration of nonlinear structures and the principle of confounded excitation, this approach provides robust resistance against machine learning attacks. Each stage of the proposed PUFs can differ and be customized according to the designer's needs. However, it must be noted that more segments may lead to lower reliability, as too complex structures may lead to the introduction of randomness in the structure. In general, a two-stage design of the proposed PUF is usually sufficient to resist modeling attacks.

5. Conclusion

In this paper, a new PUF framework is proposed. On one hand, segmentation processing is employed to introduce structural nonlinearities, thereby improving resistance against modeling attacks. On the other hand, the CXOR unit is adopted to confuse the challenges, further enhancing the resistance to machine learning attacks. The proposed framework takes resource consumption into account in its overall design, using only simple XOR logic in its confusion module to maintain the overall structure's efficiency. For oscillation-based PUFs, a new configurable ring MRO is proposed, and a two-segment MRO-MRO structure is implemented based on the proposed PUF framework. Specifically, a 128-bit MRO-MRO only costs 268 LUTs and 122 DFFs, which

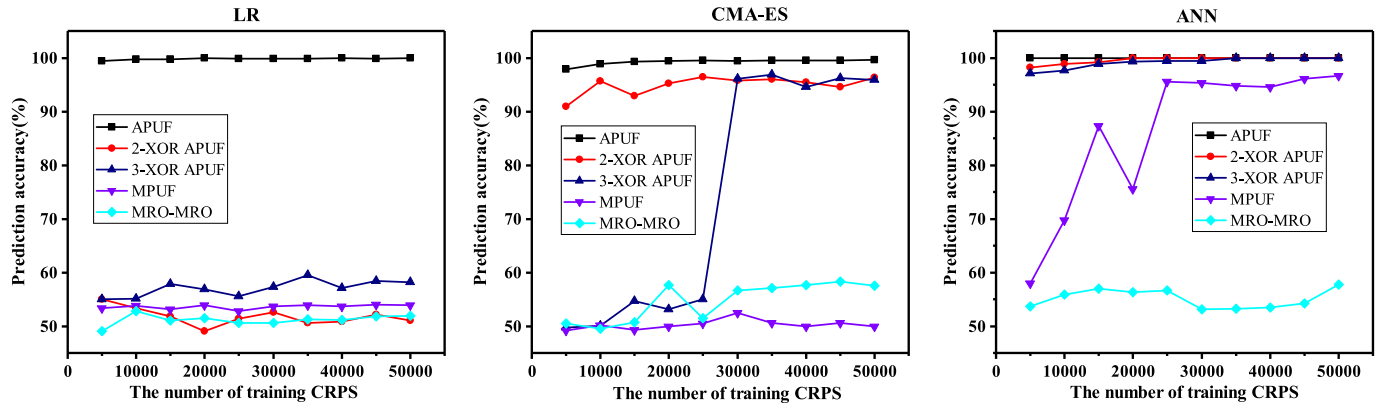


Fig. 11. MRO-MRO resistance machine learning attacks (a) LR (b) CMA-ES (c) ANN.

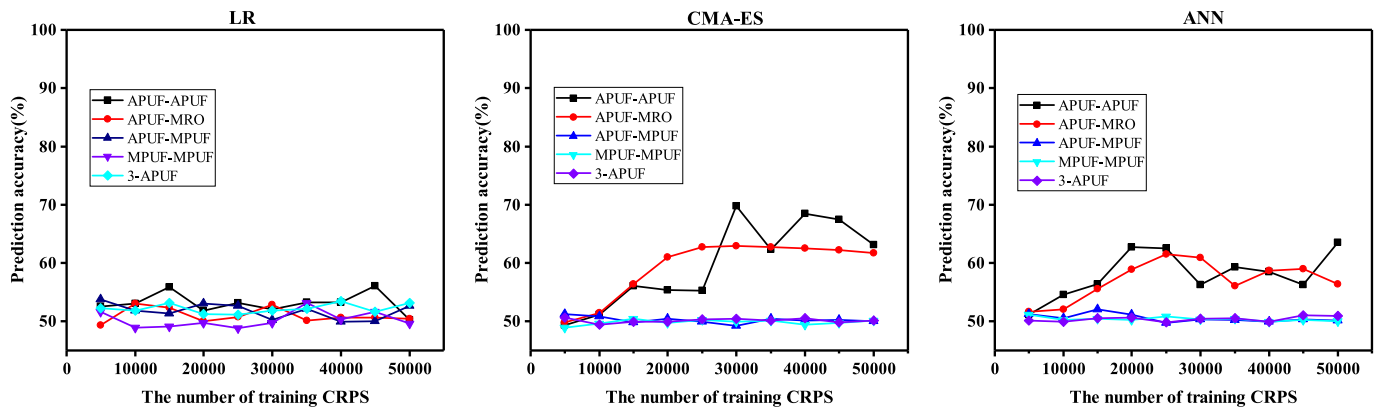


Fig. 12. Machine learning prediction rate of traditional PUF applied in the proposed PUF (a) LR (b) CMA-ES (c) ANN.

consumes fewer resources than previous works on resisting machine learning PUFs. Experiments of the MRO-MRO were conducted on two experimental platforms, Virtex-7 and Spartan-6, and the results showed that its consistency, uniqueness, and reliability were close to the ideal value. Moreover, to demonstrate that this framework enhances the resistance of classical strong PUFs against machine learning attacks, six structures, including MRO-MRO, were designed using APUF, MRO, and MPUF. Machine learning attacks were then performed using LR, CMA-ES, and ANN. The experimental results show that the resistance of the six structures to machine learning attacks is close to 50 % of the ideal value. Performance evaluation demonstrates the proposed framework is flexible, which not only can be implemented by any strong PUF, but also different segments of the PUF can be different, making it practical for industrial applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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